6 Astra Review: English Translation

Provenance. The user supplied the Chinese report on 11 October 2026 and identified the reviewer as 6 Astra. This document translates the supplied report; it is not a new review. Residual `main` citation markers are retained from the original. No model execution record or review timestamp was supplied.

My conclusion: I have not found a major mathematical gap in this version. I read the main text and Appendices A–F, checked the principal proofs, and independently recomputed the explicit example symbolically. In particular, I found no issue in the central polynomial construction theorem or the argument establishing nonmembership in $\mathcal{D}_L \cup \mathcal{D}_P$ that requires stronger core assumptions, removal of principal conclusions, or reconstruction of key proofs.

The following are the points most worth checking.

1. Fourier criterion: normalization is not omitted

A potential issue in Theorem 2.2 is that nonnegativity of the inverse Fourier transform f alone does not establish that it is a probability density.

Appendix A does address this. Writing $q = 1/M_U$, the paper first derives $q \in L^1(\mathbb{R})$ from nondegeneracy and symmetry, then proves

$$\int_{\mathbb{R}} f(x) e^{-\varepsilon x^2/2}\, dx = \frac{1}{\sqrt{2\pi\varepsilon}} \int_{\mathbb{R}} q(t) e^{-t^2/(2\varepsilon)}\, dt \longrightarrow q(0) = 1.$$

Nonnegativity of f and monotone convergence give $\int f = 1$; the argument then establishes $\widehat{f} = q$. Thus **the passage from a nonnegative inverse transform to the required probability distribution is complete**, not a skipped step. main

2. Hankel criterion: formal moments, an actual law, and the identity on the entire real axis are connected

The most vulnerable point in Theorem 2.3 would be asserting directly from recursively obtained moments that

$$\varphi_V(t) = \frac{1}{M_U(t)} \qquad (t \in \mathbb{R}).$$

The paper does not do this. Positive semidefiniteness of two families of Hankel matrices first permits construction of $X \geq 0$ through the Stieltjes moment theorem. It then sets $V = \epsilon\sqrt{X}$, where ϵ is an independent symmetric sign. Analyticity of $1/M_U$ near the origin gives

$$0 \leq v_n \leq C(2n)!\, r_0^{-2n},$$

which establishes a local exponential moment for V. main

The crucial next step considers

$$F(z) = M_U(z)\varphi_V(z) - 1.$$

This function is analytic on a horizontal strip containing the whole real axis and vanishes near the origin. The identity theorem therefore gives the reciprocal relation everywhere on the real axis. **The proof does not assume in advance that M_U is zero-free on the entire strip, so there is no circular argument.** main

The equivalence between positive semidefiniteness and strict positive definiteness is also justified: once a conjugate exists, the Fourier criterion gives an absolutely continuous law for V, under which a nonzero polynomial cannot vanish almost surely. Moment determinacy is separately proved from exponential moments. main

This establishes more than formal moment matching.

3. Polynomial construction: positivity on both sides is genuinely global

I checked this part most carefully. Theorem 3.1 must show both that the reciprocal is a characteristic function and that the original function is a moment-generating function. Its two steps address these two requirements separately. main

Step one: Laplace convolution. For

$$k = \mathcal{F}^{-1}\left[\frac{1}{P(t^2)}\right],$$

the pole structure of the rational function establishes exponential integrability of k and $\int k = 1$. Repeated poles add only polynomial factors and do not destroy exponential decay. main

The decisive uniform estimate is

$$\sup_{x\in\mathbb{R}}\left|\frac{(\ell_\lambda * k)(x)}{\ell_\lambda(x)} - 1\right| \le \int_{\mathbb{R}} |k(y)|\big(e^{\lambda|y|} - 1\big)\,dy = \delta_k(\lambda) \longrightarrow 0.$$

Consequently, when $\delta_k(\lambda) < 1$, the convolution is strictly positive for **every** $x \in \mathbb{R}$. Pointwise convergence is not mistaken for global positivity, and the spatial tails are not omitted. main

Step two: a Gaussian-polynomial density. Set $Q(s) = (1 + bs)P(s)$ and $R(y) = Q(y^2)$. Since $R(y) > 0$ and every $R^{(2j)}$ has lower degree than R,

$$C_j = \sup_{y\in\mathbb{R}} \frac{|R^{(2j)}(y)|}{R(y)} < \infty.$$

Hence

$$S_\tau(y) \ge R(y)\left(1 - \sum_{j=1}^{D} \frac{C_j}{(2\tau)^j\, j!}\right) > 0$$

for every real y when τ is sufficiently large. The appendix then verifies normalization, existence of all real exponential moments, and that the moment-generating function is precisely

$$e^{\tau t^2/2} Q(t^2).$$

The signs and scales in these derivations are correct. main

The proof thus supports the stated quantifiers

$$\exists b_0 > 0, \quad \forall b \ge b_0, \quad \exists \tau_0(b) > 0, \quad \forall \tau \ge \tau_0(b),$$

without silently requiring one τ_0 that works uniformly for all b. main

4. Excluding $\mathcal{D}_L$ and $\mathcal{D}_P$: the conditions match the external theorem

The exclusion of $\mathcal{D}_L$ is valid: the chosen P has nonreal zeros, so

$$\varphi_U(z) = e^{-\tau z^2/2}(1 - bz^2)P(-z^2)$$

has nonreal zeros, which the remaining factors do not cancel. main

For $\mathcal{D}_P$, I checked the original paper by Konstantopoulos–Patie–Sarkar. Its Theorem 4.1 and equation (4.12) indeed give

$$\log \mathcal{I}_\Psi(t) = \Phi_\Psi(t^2),$$

where Φ_Ψ is a Bernstein function. Thus the necessary condition used here,

$$\varphi_U \in \mathcal{D}_P \implies h_U(s) := \frac{1}{M_U(\sqrt{s})} \text{ is completely monotone,}$$

has the correct external basis. Institut Fourier reference.

The subsequent pole exclusion is also valid. A completely monotone function with value 1 at zero has a probability Laplace representation satisfying $|h(z)| \le 1$ on $\Re z > 0$. However,

$$h(z) = \frac{e^{-\tau z/2}}{Q(z)}$$

has nonremovable poles at nonreal zeros of Q in the closed right half-plane. Interior poles contradict analyticity; **boundary poles on the imaginary axis also contradict uniform boundedness when approached from the right**. Thus the boundary case $a_j = 0$ is not missed. main

5. Explicit example and exact threshold: recomputations agree

For $P(s) = 1 + s^2$, $b = 2$, and $\tau = 4$, I independently recomputed the partial fractions, inverse Fourier transform constants, expansion of S_4, and normalization. The results agree with the main text and appendices.

In particular,

$$S_4(y) = 2y^2 \left(y^2 - \frac{13}{8}\right)^2 + \frac{27}{32}y^2 + \frac{7}{32} > 0,$$

and the term in the other density satisfies

$$4 - \cos y + 3 \sin y \geq 4 - \sqrt{10} > 0.$$

Both are strict analytic positivity proofs, independent of numerical sampling. main

I also found no issue with the threshold $b \geq 2$ in Proposition 4.4. Sufficiency follows from products of characteristic functions and probability mixtures. Necessity follows because, for $b < 2$, the oscillating term decays more slowly than the nonoscillating positive term, forcing the inverse transform to be negative along an unbounded sequence of points. main

6. No major gap was found in the remaining results

The discrete recovery in Corollary 2.4 establishes more than finite-order moment matching: tightness, uniform integrability, and moment determinacy are handled, so the weak-convergence conclusion has a complete justification. main

The mean-value identity in Theorem 2.5 explicitly requires condition (2.9) to interchange the Taylor series and expectation. The sufficient condition for entire functions of exponential-type growth is also valid. The result is not incorrectly extended to arbitrary entire functions. main

In the recovery of known classes, convergence of the infinite independent sum and finiteness and continuity of the Gaussian variance-mixture density all have corresponding proofs. mainmain

The opinion of this review is therefore that the principal mathematical results can be retained; no major mathematical issue requiring correction has been located. This is a result-by-result mathematical review of the current manuscript, not formal verification. The external-literature check concerns the cited theorem and its applicable conditions; it does not reprove all the external papers.